آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 1

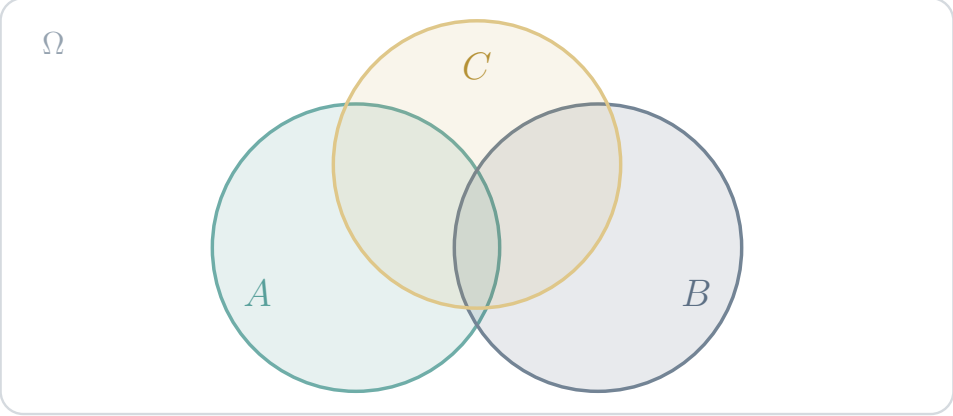
Probability

احتمال

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Learning Objectives

After completing this module, students will be able to:

1. Define random experiments and identify sample spaces for engineering scenarios
2. Apply set operations to define complex events
3. Use probability axioms to derive probability rules
4. Compute conditional probabilities
5. Determine statistical independence
6. Apply Bayes' theorem to engineering inference problems
7. Use the total probability theorem for multi-path analysis
8. Analyze random experiments with probabilistic models

1.1 Introduction: Why Probability in Engineering?

The Engineering Reality

No engineering system operates in a perfectly deterministic world. Consider:

- A digital communication system transmits bits, but **noise** corrupts some bits randomly
- A manufacturing line produces resistors, but their values **vary** around the nominal
- A sensor measures temperature, but each reading has **random error**
- A wireless signal travels to a receiver, but **fading** causes random attenuation

Probability theory provides the mathematical framework to model, analyze, and design systems that operate under uncertainty.

Engineering Question

> "If I transmit 1000 bits over a noisy channel, how many errors should I expect? Can I guarantee fewer than 5 errors?"

To answer this, we need:

1. A model of the random phenomenon (bit errors)
2. Mathematical tools to compute probabilities
3. Methods to make engineering decisions based on those probabilities

1.2 Random Experiments

Definition

A **random experiment** is a procedure that:

1. Can be repeated under identical conditions
2. Has a well-defined set of possible outcomes
3. Has an outcome that cannot be predicted with certainty before the experiment is performed

Engineering Examples

Experiment	Context	Uncertainty Source
Transmit one bit over a noisy channel	Communications	Thermal noise
Measure a resistor's value	Electronics	Manufacturing variation
Count packets arriving in 1 second	Networking	Traffic randomness
Measure received signal power	Wireless	Fading, interference
Test a component until failure	Reliability	Material defects, wear
Sample ADC output voltage	Signal processing	Quantization + noise

Key Distinction

Deterministic	Random
Output of a logic gate with known inputs	Output of a logic gate with noisy inputs
Resistance calculated from color code	Actual measured resistance
Free-space path loss at known distance	Actual received power with fading

1.3 Sample Spaces

Definition

The **sample space** S (or Ω) is the set of all possible outcomes of a random experiment.

Types of Sample Spaces

Discrete (Countable):

- Transmit one bit: $S = \{0, 1\}$ (received correctly or in error)
- Number of errors in a packet of n bits: $S = \{0, 1, 2, \ldots, n\}$
- Number of arrivals at a server: $S = \{0, 1, 2, 3, \ldots\}$

Continuous (Uncountable):

- Measured voltage: $S = \mathbb{R}$ (or a practical range like $[0, 5]V$)
- Time until component failure: $S = [0, \infty)$
- Phase of received signal: $S = [0, 2\pi)$

Engineering Example: Communication System

A transmitter sends a bit (0 or 1) over a noisy channel. The receiver makes a decision.

Sample space for (transmitted, received) pairs:

$$S = \{(0,0), (0,1), (1,0), (1,1)\}$$

	received 0	**received 1**
transmitted 0	$(0,0)$ correct	$(0,1)$ error
transmitted 1	$(1,0)$ error	$(1,1)$ correct

1.4 Events

Definition

An **event** is a subset of the sample space. An event **occurs** if the outcome of the experiment belongs to that subset.

Types of Events

- **Simple event**: Contains exactly one outcome. $E = \{(0, 1)\}$
- **Compound event**: Contains multiple outcomes. $E = \{(0, 1), (1, 0)\}$ (any error)
- **Certain event**: The entire sample space S (always occurs)
- **Impossible event**: The empty set $\emptyset$ (never occurs)

Set Operations on Events

Operation	Notation	Meaning	Engineering Interpretation
Union	$A \cup B$	A or B (or both) occurs	Either failure mode A or B triggers alarm
Intersection	$A \cap B$	Both A and B occur	Both sensors detect the signal
Complement	A^c or $\bar{A}$	A does not occur	Component does NOT fail
Difference	$A - B = A \cap B^c$	A occurs but not B	Sensor A triggers but sensor B doesn't
Mutually exclusive	$A \cap B = \emptyset$	A and B cannot both occur	Component can't be both working and failed

Engineering Example: Manufacturing Defects

A circuit board inspection checks for three types of defects:

- A = solder bridge defect
- B = missing component
- C = wrong component value

Events:

- $A \cup B \cup C$ = "board has at least one defect"
- $A \cap B$ = "board has both a solder bridge AND a missing component"
- $(A \cup B \cup C)^c$ = "board is defect-free"

1.5 Probability Axioms

Kolmogorov's Axioms

For a sample space S with events defined on it, a probability function P satisfies:

Axiom 1 (Non-negativity):

$$P(A) \geq 0 \quad \text{for any event } A$$

Axiom 2 (Normalization):

$$P(S) = 1$$

Axiom 3 (Countable Additivity):
For mutually exclusive events $A_1,\ A_2,\ A_3,\ \ldots$:

$$P(A_1 \cup A_2 \cup A_3 \cup \cdots) = P(A_1) + P(A_2) + P(A_3) + \cdots$$

Properties Derived from Axioms

Property 1: Complement Rule

$$P(A^c) = 1 - P(A)$$

Proof: $S = A \cup A^c$ and $A \cap A^c = \emptyset$, so $P(S) = P(A) + P(A^c) = 1$.

Engineering meaning: $P(\text{component works}) = 1 - P(\text{component fails})$

Property 2: Impossible Event

$$P(\emptyset) = 0$$

Property 3: Monotonicity
If $A \subseteq B$, then $P(A) \leq P(B)$

Property 4: Bound

$$0 \leq P(A) \leq 1$$

Property 5: Inclusion-Exclusion (Two Events)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Engineering meaning: When computing the probability of "failure A OR failure B," we must subtract the overlap where both fail simultaneously.

1.6 Probability Rules

Addition Rule

For any two events:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For mutually exclusive events ($A \cap B = \emptyset$):

$$P(A \cup B) = P(A) + P(B)$$

Complement Rule

$$P(A) = 1 - P(A^c)$$

This is often the easiest way to compute probabilities:

> $P(\text{at least one error in 100 bits}) = 1 - P(\text{zero errors in 100 bits})$

Inclusion-Exclusion (Three Events)

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Engineering Example: Network Redundancy

A data center has three independent network links. Each link fails with probability 0.05.

- A = link 1 fails, B = link 2 fails, C = link 3 fails
- $P(\text{at least one fails}) = P(A \cup B \cup C)$
- Using inclusion-exclusion:

$P(A \cup B \cup C) = 3(0.05) - 3(0.05)^2 + (0.05)^3 = 0.15 - 0.0075 + 0.000125 = 0.142625$
Alternatively: $P(\text{at least one fails}) = 1 - P(\text{none fail}) = 1 - (0.95)^3 = 0.142625$ ✔

1.7 Conditional Probability

Definition

The **conditional probability** of event A given that event B has occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Interpretation

Conditional probability **updates** our knowledge. Once we know B occurred, the effective sample space shrinks from S to B, and we ask: "Within B, how likely is A?"

Key Distinction

Concept	Notation	Meaning
Probability of A	$P(A)$	Likelihood of A with no additional information
Conditional probability	$P(A \mid B)$	Likelihood of A **given** that B has occurred
Joint probability	$P(A \cap B)$	Likelihood that **both** A and B occur

These are related: $P(A \cap B) = P(A \mid B) \cdot P(B) = P(B \mid A) \cdot P(A)$

Engineering Example: Detection Theory

A radar system detects aircraft:

- H_1 = aircraft is present (target)
- H_0 = aircraft is absent (no target)
- D = detector says "target present"

Important conditional probabilities:

- $P(D \mid H_1)$ = **probability of detection** (sensitivity)
- $P(D \mid H_0)$ = **probability of false alarm**
- $P(D^c \mid H_1)$ = **probability of miss** $= 1 - P(D \mid H_1)$

Suppose: $P(H_1) = 0.01$, $P(D \mid H_1) = 0.95$, $P(D \mid H_0) = 0.02$
What is $P(\text{target actually present} \mid \text{detector says present})$?
This requires **Bayes' theorem** (Section 1.9).

Engineering Example: Communication Channel

A binary symmetric channel has crossover probability $p = 0.01$:

- $P(\text{receive } 1 \mid \text{sent } 0) = p = 0.01$
- $P(\text{receive } 0 \mid \text{sent } 1) = p = 0.01$
- $P(\text{receive } 0 \mid \text{sent } 0) = 1 - p = 0.99$
- $P(\text{receive } 1 \mid \text{sent } 1) = 1 - p = 0.99$

If bits are equally likely to be 0 or 1:

- $P(\text{sent } 0 \cap \text{receive } 1) = P(\text{receive } 1 \mid \text{sent } 0) \cdot P(\text{sent } 0) = 0.01 \times 0.5 = 0.005$
- $P(\text{error}) = P(\text{sent } 0 \cap \text{receive } 1) + P(\text{sent } 1 \cap \text{receive } 0) = 0.005 + 0.005 = 0.01$

Properties of Conditional Probability

Conditional probability is itself a valid probability measure:

1. $P(A \mid B) \geq 0$
2. $P(S \mid B) = 1$
3. $P(A_1 \cup A_2 \mid B) = P(A_1 \mid B) + P(A_2 \mid B)$ if $A_1 \cap A_2 = \emptyset$

Multiplication Rule (Chain Rule)

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

For three events:

$$P(A \cap B \cap C) = P(A) \cdot P(B|A) \cdot P(C|A \cap B)$$

1.8 Independence

Definition

Events A and B are **statistically independent** if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Equivalently: $P(A \mid B) = P(A)$ — knowing B occurred gives no information about A.

Engineering Significance

Independence means **one event provides no information about the other**. This is a modeling assumption we often make:

- Noise samples at different times are independent
- Bit errors in different time slots are independent (memoryless channel)
- Component failures in different subsystems are independent

Pairwise vs. Mutual Independence

For three events A, B, C:

Pairwise independence (all three pairs):

- $P(A \cap B) = P(A)P(B)$
- $P(A \cap C) = P(A)P(C)$
- $P(B \cap C) = P(B)P(C)$

Mutual independence (all combinations):

- All pairwise conditions above, AND
- $P(A \cap B \cap C) = P(A)P(B)P(C)$

⚠ Pairwise independence does NOT imply mutual independence!

Engineering Example: Independent Component Failures

A system has three independent modules, each with failure probability 0.1.
$P(\text{all three fail}) = P(A)P(B)P(C) = (0.1)^3 = 0.001$
$P(\text{system works}) = P(\text{at least one works}) = 1 - P(\text{all fail}) = 1 - 0.001 = 0.999$
This calculation is ONLY valid because failures are independent.

Testing Independence

To verify independence from data:

1. Compute $P(A)$, $P(B)$, and $P(A \cap B)$ empirically
2. Check if $P(A \cap B) \approx P(A) \cdot P(B)$

If not approximately equal, the events are **dependent**.

Common Misconception

> ✘ "Mutually exclusive events are independent"

This is **FALSE**. If $A \cap B = \emptyset$ and $P(A) > 0,\ P(B) > 0$, then:

- $P(A \cap B) = 0$
- $P(A)P(B) > 0$
- Therefore $P(A \cap B) \neq P(A)P(B)$

Mutually exclusive events are actually **maximally dependent** — if one occurs, the other cannot.

1.9 Bayes' Theorem

Derivation

From the definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Full Statement

If $B_1, B_2, \ldots, B_n$ partition the sample space (mutually exclusive and exhaustive):

$$P(B_i|A) = \frac{P(A|B_i) \cdot P(B_i)}{\sum_{j=1}^{n} P(A|B_j) \cdot P(B_j)}$$

Terminology

Term	Name	Engineering Meaning
$P(B_i)$	Prior probability	What we believed before observing data
$P(A \mid B_i)$	Likelihood	How likely is the observation given each hypothesis
$P(B_i \mid A)$	Posterior probability	Updated belief after observing data
$P(A)$	Evidence	Total probability of the observation

Engineering Example: Fault Diagnosis

A circuit board can have a fault in one of three subsystems:

- B_1 = power supply fault, $P(B_1) = 0.3$
- B_2 = processor fault, $P(B_2) = 0.5$
- B_3 = memory fault, $P(B_3) = 0.2$

A diagnostic test produces symptom A (overheating):

- $P(A \mid B_1) = 0.8$ (power supply faults usually cause overheating)
- $P(A \mid B_2) = 0.3$ (processor faults sometimes cause overheating)
- $P(A \mid B_3) = 0.1$ (memory faults rarely cause overheating)

Question: Given that overheating is observed, what is the most likely fault?

Solution

$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + P(A \mid B_3)P(B_3)$
$= (0.8)(0.3) + (0.3)(0.5) + (0.1)(0.2)$
$= 0.24 + 0.15 + 0.02 = 0.41$
$P(B_1 \mid A) = \frac{(0.8)(0.3)}{0.41} = 0.585$ (most likely!)

$$P(B_2 \mid A) = \frac{(0.3)(0.5)}{0.41} = 0.366$$

$$P(B_3 \mid A) = \frac{(0.1)(0.2)}{0.41} = 0.049$$

Engineering Decision: Investigate the power supply first.

Engineering Example: Detection with Bayes

Returning to the radar example from Section 1.7:

- $P(H_1) = 0.01$, $P(D \mid H_1) = 0.95$, $P(D \mid H_0) = 0.02$

$P(D) = P(D \mid H_1)P(H_1) + P(D \mid H_0)P(H_0) = (0.95)(0.01) + (0.02)(0.99) = 0.0095 + 0.0198 = 0.0293$

$$P(H_1 \mid D) = \frac{(0.95)(0.01)}{0.0293} = 0.324$$

Interpretation: Even with a good detector (95% detection, 2% false alarm), when targets are rare (1%), most detections are false alarms! Only 32.4% of alarms correspond to real targets.

Engineering Lesson: System design must account for **prior probabilities**, not just detector performance.

1.10 Total Probability Theorem

Statement

If $B_1, B_2, \ldots, B_n$ form a **partition** of S (mutually exclusive, exhaustive):

$$P(A) = \sum_{i=1}^{n} P(A|B_i) \cdot P(B_i)$$

Interpretation

To find the total probability of event A, decompose into all possible "paths" to A through the partition.

Engineering Example: Multi-Path Communication

A packet can be routed through three different paths:

- Path 1: chosen 50% of the time, error probability 0.01
- Path 2: chosen 30% of the time, error probability 0.05
- Path 3: chosen 20% of the time, error probability 0.10

$P(\text{packet error}) = P(\text{error} \mid \text{path 1})P(\text{path 1}) + P(\text{error} \mid \text{path 2})P(\text{path 2}) + P(\text{error} \mid \text{path 3})P(\text{path 3})$
$= (0.01)(0.50) + (0.05)(0.30) + (0.10)(0.20)$
$= 0.005 + 0.015 + 0.020 = 0.040$
Overall error rate: 4%

Engineering Example: Manufacturing with Multiple Suppliers

A company sources capacitors from two suppliers:

- Supplier A: provides 60% of capacitors, defect rate 2%
- Supplier B: provides 40% of capacitors, defect rate 5%

$P(\text{defective}) = P(\text{defective} \mid A)P(A) + P(\text{defective} \mid B)P(B)$
$= (0.02)(0.60) + (0.05)(0.40)$
$= 0.012 + 0.020 = 0.032$
If a defective capacitor is found, which supplier likely produced it?
$P(A \mid \text{defective}) = \dfrac{(0.02)(0.60)}{0.032} = 0.375$
$P(B \mid \text{defective}) = \dfrac{(0.05)(0.40)}{0.032} = 0.625$
Despite providing fewer units, Supplier B is more likely responsible for a given defect.

1.11 Summary: Key Distinctions

Concept	Definition	Engineering Meaning
Event	A subset of the sample space	A specific outcome or set of outcomes we care about
Probability	$P(A)$ = measure of likelihood	Long-run relative frequency of occurrence
Conditional Probability	$P(A \mid B)$ = updated probability given B	How knowledge of one event changes another's likelihood
Independence	$P(A \cap B) = P(A)P(B)$	Knowing one event tells nothing about the other

1.12 Practice Problems

Problem 1: Component Reliability

A system has two independent processors. Processor 1 fails with probability 0.02, Processor 2 fails with probability 0.03. The system fails only if BOTH processors fail.

(a) What is the probability that the system fails?
(b) What is the probability that at least one processor fails?
(c) Given that the system is still working, what is the probability that processor 1 has failed?

Solution

(a) $P(\text{system fails}) = P(\text{both fail}) = (0.02)(0.03) = 0.0006$
(b) $P(\text{at least one fails}) = 1 - P(\text{none fail}) = 1 - (0.98)(0.97) = 1 - 0.9506 = 0.0494$
(c) System working = NOT(both fail) = at least one works.

$$P(P1 \text{ failed} \mid \text{system works}) = \frac{P(P1 \text{ failed} \cap P2 \text{ works})}{P(\text{system works})}$$

$$= \frac{(0.02)(0.97)}{1 - 0.0006} = \frac{0.0194}{0.9994} \approx 0.0194$$

Problem 2: Sensor Detection

A fire detection system has: $P(\text{fire}) = 0.001$, $P(\text{alarm} \mid \text{fire}) = 0.98$, $P(\text{alarm} \mid \text{no fire}) = 0.01$.

(a) What is $P(\text{alarm})$?
(b) If the alarm sounds, what is $P(\text{fire} \mid \text{alarm})$?
(c) Comment on the result.

Solution

(a) $P(\text{alarm}) = P(\text{alarm} \mid \text{fire})P(\text{fire}) + P(\text{alarm} \mid \text{no fire})P(\text{no fire})$
$= (0.98)(0.001) + (0.01)(0.999) = 0.000980 + 0.009990 = 0.010970$

(b) $$P(\text{fire} \mid \text{alarm}) = \frac{P(\text{alarm} \mid \text{fire})P(\text{fire})}{P(\text{alarm})}$$
$$= \frac{(0.98)(0.001)}{0.01097} = 0.0894$$

(c) Only $\sim 9\%$ of alarms correspond to real fires! This is the base rate fallacy. Even with an excellent detector, rare events produce many false alarms.

Problem 3: Communication Error

A packet of 8 bits is transmitted. Each bit has independent error probability 0.05.

(a) What is $P(\text{no errors in the packet})$?
(b) What is $P(\text{exactly one error})$?
(c) What is $P(\text{at least one error})$?

Solution

(a) $P(\text{no errors}) = (0.95)^8 = 0.6634$
(b) $P(\text{exactly one error}) = C(8,\,1)(0.05)^1(0.95)^7 = 8 \times 0.05 \times 0.6983 = 0.2793$
(c) $P(\text{at least one error}) = 1 - P(\text{no errors}) = 1 - 0.6634 = 0.3366$

Problem 4: Total Probability Application

A wireless system operates in three modes:

- Mode 1 (good channel): 60% of time, $P(\text{outage}) = 0.01$
- Mode 2 (moderate channel): 30% of time, $P(\text{outage}) = 0.10$
- Mode 3 (poor channel): 10% of time, $P(\text{outage}) = 0.50$

(a) What is the overall outage probability?
(b) Given an outage occurred, what is the probability the system was in Mode 3?

Solution

(a) $P(\text{outage}) = (0.01)(0.60) + (0.10)(0.30) + (0.50)(0.10) = 0.006 + 0.030 + 0.050 = 0.086$
(b) $P(\text{Mode 3} \mid \text{outage}) = \dfrac{(0.50)(0.10)}{0.086} = 0.581$

Problem 5: Independence Analysis

Two sensors monitor a process. Sensor A has false alarm probability 0.05. Sensor B has false alarm probability 0.03.

(a) If sensors are independent, what is $P(\text{both give false alarm})$?
(b) If sensors share a noise source and $P(\text{both false alarm}) = 0.01$, are they independent?
(c) What is $P(A \text{ false alarm} \mid B \text{ false alarm})$ in case (b)?

Solution

(a) If independent: $P(\text{both}) = (0.05)(0.03) = 0.0015$
(b) If $P(\text{both}) = 0.01 \neq 0.0015$, they are NOT independent. The shared noise source creates dependence.
(c) $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.01}{0.03} = 0.333$

This is much higher than $P(A) = 0.05$, confirming strong dependence.

1.13 Key Takeaways

1. **Probability quantifies uncertainty** — essential for engineering design under uncertainty
2. **Conditional probability updates beliefs** — the foundation of detection, estimation, and diagnosis
3. **Independence simplifies analysis** — but must be justified, not assumed
4. **Bayes' theorem inverts conditional probabilities** — from "likelihood of evidence given cause" to "likelihood of cause given evidence"
5. **Base rates matter** — even good detectors produce false alarms when events are rare

Next Module: Module 2 — One Random Variable and Its Functions